



Answer the following FOUR questions:

(Total Marks: 50)

1st Question

Marks: 15

For each of the following statements correct the underlined term.

1. Requirements elicitation is the process of understanding and controlling changes to requirements. *analysis & simulation* **management change**
2. Batch proceeding systems are developed by scientists and engineers to model physical situations. **system modeling and simulation**
3. Agile approach is suitable for developing a software when you have a team with relatively low skill levels. *plan driven* **plan Driven**
4. In incremental development, the first version of the software is available for end users. *waterfall model* **deliver**
5. Users of the system shall authenticate themselves using their authority identity card is an example of external requirement. *org. req* **reuse a component analysis**
6. Component analysis is an essential phase in waterfall software process. *reuse* **refactroing**
7. Collective ownership means that all developers are expected to improve the code and keep the code simple and maintainable.
8. Availability is a quality factor that means that the system should be easy to use and organized in such a way that user errors are minimized.
9. Data store in data flow diagrams represent the origin or destination of data (outside the system). *entity*
10. Lack of ambiguity provides an indication of the stability of a software product.
11. Change tolerance is where the software process includes activities that can anticipate possible changes before significant rework is required. *avoidance* **prototyping**
12. Error checking and recovery may not be included in the system increment because it is an initial version of the system and focuses on areas of the product that are not well-understood.
13. Project manager is responsible for prioritizing the product backlog.
14. Error flow is the "happy day" scenario, or the most frequently used path through the use case. *primary*
15. Development testing is where a separate testing team tests a complete version of the system before it is released to users.

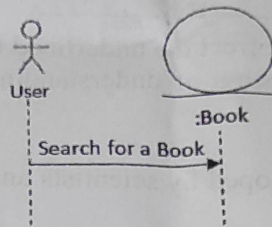
2nd Question

Marks: 10

A) Explain two reasons why in practice, the principles underlying agile methods are sometimes difficult to realize. *bec Time is short* *bec this is not a goal & it's a requirement* *incremental driven* **time pressure and deadline** **skile teams requirement** [3 Marks]

B) Mention Scrum framework roles. [3 Marks]

C) The following diagram is incorrect. Mention why and redraw it correctly. [4 Marks]



boundary

entity

3rd Question

Marks: 10

int a = 5, b = 7, c = 0; ①

while (a <= 10) ②

{

 If (a==b)

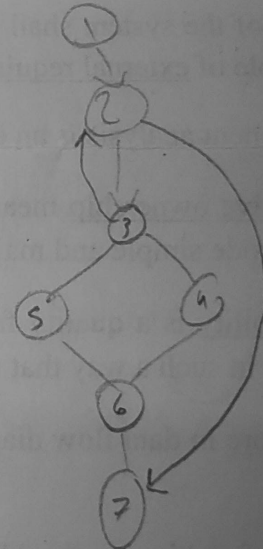
 { c=c+b; }

 Else {c=c+1; }

 a = a + 1;

}

Console.WriteLine(c);



Given the above code, answer the following questions:

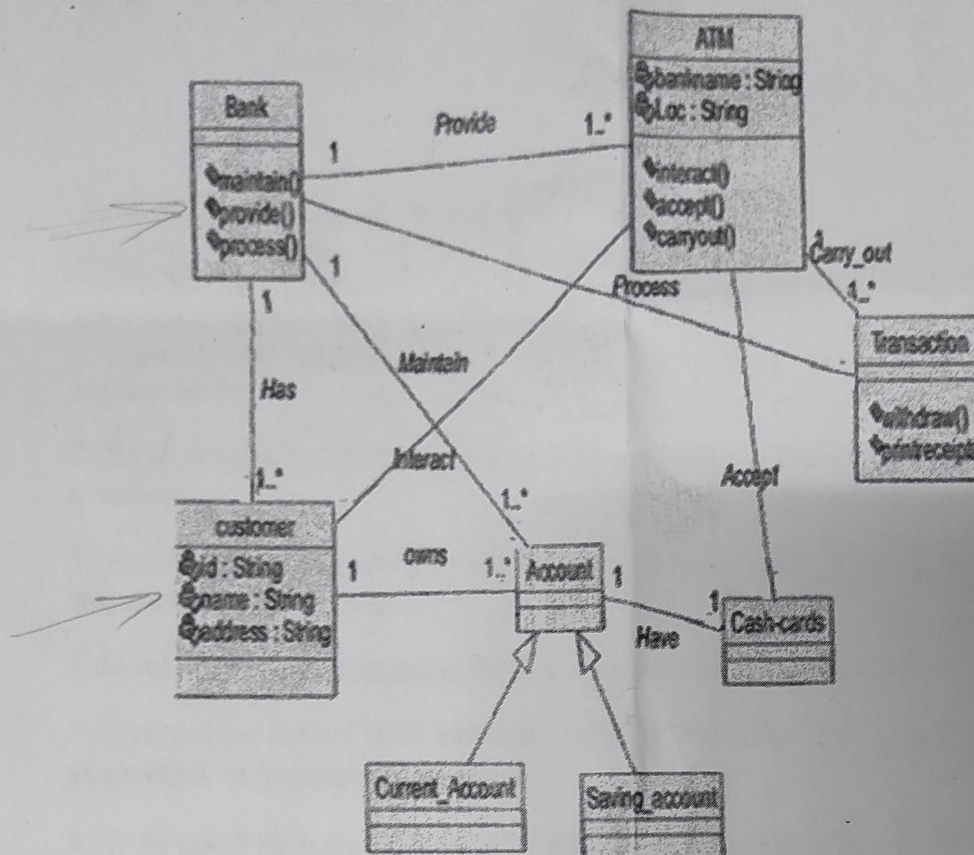
1. Draw the corresponding flow graph. [2 Marks]

2. Calculate the Cyclomatic Complexity using two methods. 3 [2 Marks]

3. Write the basis set of linearly independent paths. 3¹ [2 Marks]

4. Prepare test cases that will force execution of each path in the basis set. [2 Marks]

5. Find the overall program length and volume. [2 Marks]



- A) Using the previous class diagram, answer the following questions: [5 Marks]
- 1) What are id, name, and address in class customer called?
 - 2) What are maintain, provide, and process in class bank called?
 - 3) Mention a generalization relationship.
 - 4) Explain the multiplicity of provide relationship between bank and ATM.
 - 5) Give the suitable multiplicity for interact relationship between customer and ATM.

- B) Explain the following terms in the data flow diagram [5 Marks]
- 1) Process
 - 2) Data flow
 - 3) External entity
 - 4) Data store
 - 5) Context diagram

- C) Give brief comparison between: [5 Marks]
- 1) Unit testing and integration testing
 - 2) White box and Black box testing

$n = 4$
20